

1. Unified big data Architecture for a multinational retail organization

Introduction

A multinational retail company has customer data from different sources like online shopping, mobile apps, POS systems and loyalty cards. This creates fragmented data.

Main components.

1. Data sources - POS, website, mobile app, CRM and social media
2. Data ingestion - kafka collects real-time data
3. Data Lake - Hadoop/cloud storage stores large amount of data.
4. Processing - Apache spark processes the data
5. Customer profile - Combines all customer activities into one profile.
6. Recommendation system - ML analyzes customer behaviour and suggests products.

Benefits.

- * Single view of customer
- * Real-time analytics
- * Personalized recommendations
- * Better customer experience
- * Scalable for large data.

Conclusion.

A unified big data architecture removes data fragmentation and helps the retailer provide real-time and personalized services.

2. Distributed Computing for scientific research.

Scientific research produces huge amount of data. A centralized system takes more time to process this data.

Distributed computing

- * Data is divided into smaller parts
- * Each computer/node processes one part
- * All nodes work simultaneously
- * Results are combined at the end

Parallel processing Advantages.

1. High speed - Many tasks run simultaneously.
 2. Scalability - More nodes can be added.
 3. Fault tolerance - Failure of one node does not stop the complete system.
 4. Cost effective - Uses multiple normal computers.
 5. Large data processing - Suitable for scientific data.
- example: Hadoop and Apache spark.

Conclusion:

Distributed and parallel computing is more scalable than centralized computing for large scientific datasets.

3. Fraud detection in digital payments.

Digital payment systems generate huge numbers of transactions. Big data analytics can detect fraudulent transactions quickly.

Steps.

1. Collect transaction amount, location, device and user details.
2. Use kafka to collect real-time transaction data.
3. Use spark streaming for real-time processing.
4. Extract feature like unusual amount, location and repeated transactions.
5. Use ML models to identify fraud.
6. Generate a risk score.
7. Block high-risk transactions and allow normal transactions.

Benefits.

- * Real time fraud detection
- * Reduces financial loss
- * Detects unusual patterns
- * Faster response
- * Improves accuracy.

Conclusion.

Streaming processing and predictive models help detect and prevent fraud before major damage occurs.

4. Multimedia data in Big data.

Multimedia data includes images, audio and videos. Managing the data is difficult because of its large size and different formats. Challenges.

1. Huge storage requirement.
2. High processing time
3. Different file formats

Suitable technologies.

- * cloud object storage
- * HDFS
- * NO SQL

5. Disaster recovery and business continuity.

A cloud big data system can fail because of hardware failure, network problems, cyberattacks or natural disasters. Disaster recovery keeps the system running.

Important methods.

1. Redundancy
2. Replication
3. Backup
4. Automated Failover
5. Load balancing
6. Monitoring
7. RTO
8. RPO

It makes cloud big data systems reliable and resilient.